

GED Math Practice: Ratios, Proportions, and Percents

November 6, 2025

Instructions

Show your work. Calculators are allowed. Unless stated otherwise, round money to the nearest cent and other answers to the nearest tenth.

Practice Questions (20)

1. Simplify the ratio 18 : 24.
2. The ratio of cats to dogs at a shelter is 12 to 15. Write this ratio in simplest form.
3. A class has 14 girls and 11 boys. Write the ratio of boys to total students in simplest form.
4. Convert the ratio 150 cm : 2 m to simplest form.
5. A recipe uses 3 cups of flour and 2 cups of sugar. If you use 7.5 cups of flour, how much sugar keeps the ratio the same?
6. Solve the proportion: $\frac{x}{12} = \frac{5}{8}$.
7. Solve the proportion: $\frac{14}{x} = \frac{7}{9}$.
8. On a map, 1 cm represents 8 km. How many kilometers are represented by 3.6 cm?
9. A car travels 210 miles on 7 gallons of gas. At the same rate, how far can it travel on 10 gallons?
10. A store mixes nuts in a ratio of 2 parts almonds to 3 parts peanuts. If the final mix weighs 10 lb, how many pounds are almonds?
11. Find 35% of 480.
12. 54 is what percent of 90?
13. 27 is 15% of what number?
14. An item increases in price from \$64 to \$80. Find the percent increase.

15. A jacket marked \$120 is discounted 25%. What is the sale price?
16. A meal costs \$48 before tax. Sales tax is 13% and you tip 15% on the pre-tax amount. What is the total cost?
17. Simple interest: You invest \$600 at an annual simple interest rate of 4% for 3 years. How much interest is earned, and what is the final amount?
18. A population drops from 1,250 to 1,100. Find the percent decrease.
19. In a survey, 120 people prefer tea and 180 prefer coffee. What percent prefer tea?
20. Multi-step: A laptop has a list price of \$950. It goes on sale for 20% off, then a coupon takes an additional 10% off the sale price. After discounts, 13% tax is added. What is the final price?

Full Solutions

1. **Solution.** $18 : 24 = \frac{18}{24} = \frac{3}{4}$ after dividing by 6. So 3 : 4. □
2. **Solution.** $12 : 15 = \frac{12}{15} = \frac{4}{5}$ after dividing by 3. So 4 : 5. □
3. **Solution.** Total = $14 + 11 = 25$. Boys:total = 11 : 25 (already simplest). □
4. **Solution.** 2 m = 200 cm. So $150 : 200 = \frac{150}{200} = \frac{3}{4}$, i.e., 3 : 4. □
5. **Solution.** Flour:sugar = 3 : 2 = $\frac{3}{2}$. Let sugar = x . Set $\frac{3}{2} = \frac{7.5}{x} \Rightarrow 3x = 15 \Rightarrow x = 5$. Use 5 cups sugar. □
6. **Solution.** $\frac{x}{12} = \frac{5}{8} \Rightarrow 8x = 60 \Rightarrow x = 7.5$. □
7. **Solution.** $\frac{14}{x} = \frac{7}{9} \Rightarrow 7x = 126 \Rightarrow x = 18$. □
8. **Solution.** Scale 1 cm : 8 km. So 3.6 cm $\rightarrow 3.6 \times 8 = 28.8$ km. □
9. **Solution.** Unit rate = $210/7 = 30$ mpg. For 10 gal: $30 \times 10 = 300$ miles. □
10. **Solution.** Total parts = $2 + 3 = 5$. Almonds = $\frac{2}{5} \times 10 = 4$ lb. □
11. **Solution.** $0.35 \times 480 = 168$. □
12. **Solution.** $\frac{54}{90} = 0.6 = 60\%$. □
13. **Solution.** $27 = 0.15 N \Rightarrow N = \frac{27}{0.15} = 180$. □
14. **Solution.** Change = $80 - 64 = 16$. Percent increase = $\frac{16}{64} = 0.25 = 25\%$. □
15. **Solution.** Discount = $0.25 \times 120 = 30$. Sale price = $120 - 30 = \$90.00$. □
16. **Solution.** Tax = $0.13 \times 48 = \$6.24$. Tip (on pre-tax) = $0.15 \times 48 = \$7.20$. Total = $48 + 6.24 + 7.20 = \$61.44$. □
17. **Solution.** Simple interest $I = Prt = 600 \times 0.04 \times 3 = \72 . Final amount = $600 + 72 = \$672$. □
18. **Solution.** Change = $1,250 - 1,100 = 150$. Percent decrease = $\frac{150}{1,250} = 0.12 = 12\%$. □
19. **Solution.** Total = $120 + 180 = 300$. Percent tea = $\frac{120}{300} = 0.4 = 40\%$. □
20. **Solution.** First discount: 20% of 950 is 190. Sale price = $950 - 190 = 760$. Second discount: 10% of 760 is 76. New price = $760 - 76 = 684$. Tax: 13% of 684 is 88.92. Final price = $684 + 88.92 = \$772.92$. □